

Evaluating cluster detection algorithms and feature extraction techniques in automatic classification of fish species

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Abstract This paper proposes five different schemes for automatic classification of fish species. These schemes make the species recognition based on image sample analysis. Different techniques have been combined for building the classifiers: three feature extraction techniques (PCA, SIFT and SIFT + VLAD + PCA), three data clustering algorithms (aiNet, ARIA and *k*-means) and three input classifiers (*k*-NN, SIFT class. and *k*-means class) are tested. When compared to common methodologies, which are based on human observation, it is believed that these schemes are able to provide significant improvement in time and financial resources spent in classification. Two datasets have been considered: (1) a dataset with image samples of six fish species which are perfectly conserved in formaldehyde solution, and; (2) a dataset composed of images of four fish species in real-world conditions (in vivo). The five proposed schemes have been evaluated

in both datasets, and a ranking for the methods has been derived for each one.

Keywords Fish automatic classification · Feature extraction · Image clustering

1 Introduction

The information about fish distribution and abundance can be very valuable to several practical applications, such as feeding strategies in fish farms [7], design of fish ladders in hydroelectric generation facilities [12], ecological/environmental studies [4] and stock assessment for fishery management [14]. An adequate stock assessment is crucial for fisheries management, especially because human demand for fish is continuously increasing [9]. It is necessary to match the natural stock fluctuation and the fishing efforts to avoid irrecoverable damage to the exploited species. Long-term consequences of over-fishing can be drastic, since fish resources provide vital contributions to food supplies and they are strictly related with employment in coastal areas.

Recently, several techniques have been proposed for estimating the fish availability to avoid unfair exploitation of fish resources. These methods, although useful, are hardly dependent on the quality of the information available. Virtual population analysis (VPA) methods, for instance, make use of commercial information, which can introduce considerable bias in stock evaluation [9, 17]. Accurate predictions are only possible when large amount of unbiased data is available, and, therefore, the strength of any stock biomass prediction is strictly related with the quality of the available inputs [14, 28].

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A very important task which must be performed during information acquisition is the fish species classification. This classification procedure can be seen as the task of grouping and categorizing fish species according to their shared characteristics. Human classification of fish species is completely unviable for practical situations, since it often implies in big financial investments and it is highly susceptible to human errors.

To overcome the problems related with human classification, some recent works have proposed automatic methods for classifying fish species [6, 19, 31, 33]. A significant part of these methods is vision based and, according to Nery et al. [25], they should be able to handle with conditions just like:

- arbitrary scale and orientation: fish appear in a variety of scales, orientations and body poses;
- environmental variations: the illumination and water transparency may vary;
- bad image quality: image acquisition is frequently affected by noise and distortions in the optical system;
- segmentation failures: the segmentation of an individual fish may not be reliable.

This paper proposes a set of novel vision-based algorithms driven to automatic classification of fish species. All the proposed algorithms share a similar structure (Fig. 1), with two separated modules: (1) knowledge building module and, (2) classification module. In the diagram of Fig. 1, each block can be understood as:

Feature extraction: in this block, features of an input image are extracted to build an adequate description of the image. (1) Principal component analysis (PCA) [27], (2) scale-invariant feature transform (SIFT) [20], and (3) SIFT + vector of locally aggregated descriptors (VLAD) + PCA [15] feature extraction techniques are evaluated here.

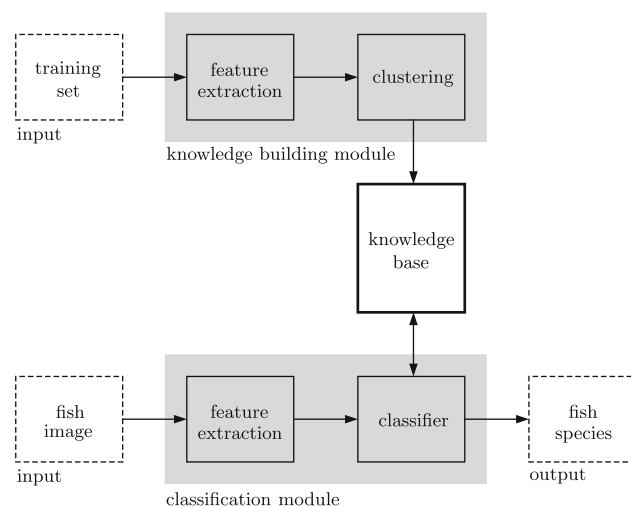


Fig. 1 The proposed framework

Clustering: the role of the clustering block is to create groups from the individuals of the training set, based on their similar characteristics. In the proposed classification schemes, this task can be performed by three different algorithms: (1) artificial immune network (aiNet) [10], (2) adaptive radius immune algorithm (ARIA) [3] and, (3) k -means algorithm [24].

Classifier: the intention of the classifier block is to detect which group shares more similarities with an input image to determine which species is more suitable for such an input. Here, this task is performed by: (1) a k -nearest neighbor classifier (k -NN) [26], (2) a SIFT classifier (classifier adapted to work with descriptors generated by SIFT and, (3) a k -means based classifier, which exploits the characteristics of k -means for faster inspection of the knowledge database.

The main idea of these classification schemes is to propose flexible and general tools for performing classification of different fish species. An evaluation of these methods is also performed here to find which one is the most suitable for a particular dataset.

The paper is structured as follows:

- The related work is presented in Sect. 2.
- The five classification schemes which are proposed here for automatic fish classification are shown in Sect. 3. All the techniques which are used inside these schemes are also described in this section.
- The datasets which are employed for evaluating the proposed schemes are discussed in Sect. 4
- Numerical results for two datasets are finally presented in Sect. 5.

2 Related work

Although automatic classification of fish species can be very convenient for real applications, the set of works which proposed really effective solutions for that problem is still limited [2, 5, 6, 9, 19, 25, 30, 31, 33]. A brief description of such works is given below.

Cadieux et al. [6], uses an infrared silhouette sensor to observe contours of fish in constrained flow. Classification is performed based on a committee of three different classifiers, which perform fish silhouette recognition using invariant moments and Fourier boundary descriptors. Those features do not work well for noisy images, what can be restrictive in real cases, due to environmental variations. The authors reported a classification accuracy close to 78 %.

Lee et al. [19] proposes a shape analysis for removing edge noise and redundant data points. A curvature function analysis is used to locate critical landmark points, which

are used to extract the fish contour segments for species classification. The authors performed experiments with a small dataset, composed of 22 sample images.

In [25], the authors propose a feature selection methodology for fish species classification. Given a set of available descriptors, the method builds the feature vector by estimating the contribution of each individual characteristic to the overall classification performance. Two statistical indicators, so-called discrimination and independence, are used to support the feature selection process. The authors reported a classification accuracy of 85 %.

In [33], the authors present a deformable template object recognition method for classifying two fish species in underwater video. A deformable template matching, which employs shape contexts and large-scale spatial structure preservation, is adopted. A support vector machine (SVM) is used for classifying texture, and it reached 90 % of classification accuracy.

Bermejo et al. [2] proposes a vision-based procedure for evaluating fish age using otolith images. Morphological and statistical feature extraction (PCA) procedures are used jointly with a multi-class SVM, which intends to group the fishes into classes, based on their ages. A mean accuracy of 73 % was observed in the best case.

In [31], the authors propose a system for classifying four fish species. In this system, the feature extraction is performed using PCA and a aiNet algorithm is employed for clustering the training inputs. An overall accuracy level higher than 80% is reported by the authors. This work, jointly with [32], can be considered as preliminary studies of the present work.

Cabreira et al. [5] employ an Artificial Neural Network (ANN) for classifying digital echo recordings of fish schools. Energetic, morphometric and bathymetric descriptors are extracted. Several fish species have been considered, including anchovy, rough scad and blue whiting. Correct classification rates up to 96 % were obtained. That method is best suited to coarser detections, such as finding fish schools, and cannot be applied in multispecies environments characterized by continuous changes in the species composition of the schools.

The work [9] also employs acoustic descriptors, ANN and Discriminant Function Analysis (DFA) for categorizing fish schools. Experiments with three different schools have been conducted and an accuracy level of 87 % has been reached. This approach shares the same limitations of the one introduced in [5].

Robothama et al. [30] propose an algorithm for identifying small fish species, based on acoustic data. Four categories of descriptors have been considered (morphological, bathymetric, energetic and positional) and the classification is performed using SVM or multi-layer

perceptron (MLP) neural networks. Correct classification rates up to 89 % are observed by the authors.

When compared to the works described above, the set of tools proposed here tend to be more flexible, since it does not depend on problem-specific descriptors. The availability of multiple techniques for feature extraction, clustering and classification makes possible to look for a method which is better suited for any particular application, including those ones which are not related to fish classification.

The works aforementioned are directly associated with fish classification problems. However, semi-supervised classification methods have recently proved to be quite efficient and can be used in works related to classification problems [38, 42]. Thus, considering the relevance of these methods in pattern classification a brief description of these studies is given as follows.

In [39], the authors present a pairwise constraints based multiview subspace learning (PC-MSL) method for real applications in scene classification. The method proposes to solve the problem of multiview dimensionality reduction by learning a unified low-dimensional subspace to effectively fuse the multiview features. They conducted experiments on two datasets: a dataset of natural and indoor scenes and compared PC-MSL with the performance of eight different methods. Results demonstrated the effectiveness of the proposed method in all cases.

Yu et al. [40] propose an adaptive hypergraph learning algorithm for transductive image classification. The approach not only investigates a robust hyperedge construction method but also presents a simultaneous learning of the labels of unlabeled images and the weights of hyperedges. In the hypergraph construction, a hyperedge to link an image with its varying-size neighborhood has been generated. In the learning process, an alternating optimization method is introduced to optimize both the labels and hyperedge weights. For the experiments of classification, they compared the performance of ten methods and the results demonstrated the effectiveness of the proposed method in all cases.

In [41], the authors propose a semi-supervised multiview distance metric learning (SSM-DML) for cartoon synthesis. Based on graph-based semi-supervised learning, SSM-DML learns the multiview distance metrics from multiple feature sets and from the labels of unlabeled cartoon characters simultaneously. SSM-DML discovers complementary characteristics of different feature sets through an alternating optimization-based iterative algorithm. The authors compare the effectiveness of the proposed SSM-DML with five different methods in the multiview cartoon character classification (multi-CCC) module.

In [22] and [23], the authors present a multiview vector-valued manifold regularization (MV³MR) for multi-label

image classification in which images are naturally characterized by multiple views. MV³MR exploits the complementary property of different features and discovers the intrinsic local geometry of the compact support shared by different features under the theme of manifold regularization. The authors performed intensive experiments on two challenging datasets PASCAL VOC'07 and MIR Flickr showing that MV³MR outperforms the traditional multi-label algorithms as well as some well-known multiple kernel learning methods.

Luo et al. [21] propose a manifold regularized multitask learning (MRMTL) algorithm for semi-supervised multi-label image classification. MRMTL learns a discriminative subspace shared by multiple classification tasks by exploiting the common structure of these tasks. It effectively controls the model complexity because different tasks limit one another's search volume, and the manifold regularization ensures that the functions in the shared hypothesis space are smooth along the data manifold. The authors conducted extensive experiments, on the PASCAL VOC'07 dataset with 20 classes and the MIR dataset with 38 classes, by comparing MRMTL with popular image classification algorithms.

In [37], the authors propose an online NMF (ONMF) algorithm to efficiently handle very large-scale and/or streaming datasets. Unlike conventional NMF [18] solutions which require the entire data matrix to reside in the memory, ONMF algorithm proceeds with one data point or one chunk of data points at a time. Experiments demonstrated that even with only one pass of the data, the ONMF algorithm can achieve nearly the same performance as the conventional NMF method.

Finally, Guan et al. [13] propose an efficient online nonnegative matrix factorization with robust stochastic approximation algorithm (OR-NMF) to learn nonnegative matrix factorization (NMF) from large-scale or streaming datasets. In particular, the authors treated NMF as a stochastic optimization problem and utilized the robust stochastic approximation method (RSA) to update the basis matrix in an online fashion. The experimental results of face recognition and image annotation on public datasets confirmed that the performance of OR-NMF is superior to other online NMF (ONMF) algorithms.

3 Classification schemes

In this paper, five classification schemes are proposed for automatic fish classification. All these schemes have the same structure, which is shown in Fig. 1. In practice, the five schemes vary only on the techniques which are used for feature extraction, clustering and classification, such as shown in Table 1.

Table 1 Scheme components

	Feature extraction	Clustering	Classifier
Scheme 1	PCA	aiNet	<i>k</i> -NN
Scheme 2	PCA	ARIA	<i>k</i> -NN
Scheme 3	SIFT	aiNet	SIFT class
Scheme 4	SIFT	ARIA	SIFT class
Scheme 5	SIFT + VLAD + PCA	<i>k</i> -means	<i>k</i> -means based class

Note that using the techniques proposed in Table 1, there are 15 applicable combinations, considering that: (1) the SIFT classifier proposed should be used only when the corresponding SIFT feature extraction method is applied and (2) similarly, the *k*-means based classifier should be used only when the *k*-means clustering properties are exploited.

That said, it is also important to consider that the three feature extraction methods proposed in this work represent three different scenarios, specifically: (1) one based on a global image description, in this case, using the well-known PCA method, (2) another one based on local image descriptors, built from interesting points computed by the SIFT method, and (3) finally, one based on a mix that involves global and local image description, where the VLAD method is jointly used with SIFT and PCA.

Therefore, given those three specific scenarios, this work focusses the investigation on the five classification schemes in Table 1 (that is, one third of the 15 applicable combinations), since they: (1) on the one hand (schemes 1 to 4) represent the combination of classical approaches for feature extraction (PCA and SIFT) with promising clustering techniques (artificial immune systems, such as, aiNet and ARIA) that still demand on deeper studies to demonstrate their applicability and limitations, and (2) on the other hand (scheme 5) demonstrate the use of a state-of-the-art approach for feature extraction (SIFT + VLAD + PCA) with one of the simplest unsupervised learning algorithms to classify a given data set through a certain number of clusters fixed a priori *k*-means. All the techniques cited in Table 1 are described in the remainder of this section.

3.1 Feature extraction

Local feature extraction and representation are recognized as critical tasks inside fish species classification algorithms [33]. Here, three different techniques are evaluated: principal component analysis (PCA) [27], scale-invariant feature transform (SIFT) [20] and SIFT + vector of locally aggregated descriptors (VLAD) + PCA [15]. These techniques are briefly discussed below.

3.1.1 Principal component analysis (PCA)

Principal component analysis is a statistical technique commonly employed for reducing space dimension. Although PCA presents some shortcomings, such as its implicit assumption of Gaussian distributions and its restriction to orthogonal linear combinations, it remains popular due to its simplicity [16]. A description of how PCA can be used for object classification is given below.

Initially, the *RGB* (red green blue) image samples are converted to *YUV* color space (*Y*: brightness, *U* and *V*: color). This conversion is necessary because *YUV* color space provides better modeling of human perception than *RGB* color space.

Given p image samples, three n -dimensional column vectors $\mathbf{y}_i$, $\mathbf{u}_i$ and $\mathbf{v}_i$ are obtained for each sample i ($i = 1, 2, \dots, p$), by concatenating the n pixel values of each component *Y*, *U* and *V*. Those n -dimensional column vectors are combined to form three different matrices:

$$\Psi = [\mathbf{y}_1 \mathbf{y}_2 \dots \mathbf{y}_p], \quad (1)$$

$$\mathcal{T} = [\mathbf{u}_1 \mathbf{u}_2 \dots \mathbf{u}_p], \quad (2)$$

$$\varsigma = [\mathbf{v}_1 \mathbf{v}_2 \dots \mathbf{v}_p], \quad (3)$$

which encode the brightness (Ψ) and color ($\mathcal{T}$ and ς) information of all image samples.

In the remainder of this section, the application of PCA to matrix $\mathcal{T}$ is described. A similar procedure is also applied to matrices Ψ and ς .

Consider a new coordinate system $T = [\mathbf{t}_1 \mathbf{t}_2 \dots \mathbf{t}_p]$. Supposing that T is orthonormal, the representation $\hat{\mathcal{T}}$ of $\mathcal{T}$ in this new system is given by:

$$\hat{\mathcal{T}} = T^\top \mathcal{T}. \quad (4)$$

Assuming that $\mathbf{u}_i$ has expected value zero, that is, $E[\mathbf{u}_i] = \mathbf{0}, \forall i$, we compute the covariance matrix of $\hat{\mathcal{T}}$ as follows:

$$\sigma^2 = T^\top R T, \quad (5)$$

in which $R = E[\mathcal{T}\mathcal{T}^\top] \simeq \frac{1}{p} \mathcal{T}\mathcal{T}^\top$.

The coordinate system T which results in the highest possible value for covariance is computed by finding a singular value decomposition for R , as follows:

$$R = \Lambda \Sigma \Omega^\top. \quad (6)$$

Since R is symmetric, $\Lambda = \Omega$, which is $R = \Omega \Sigma \Omega^\top$. After some mathematical manipulation:

$$\Sigma = \Omega^\top R \Omega, \quad (7)$$

in which the main diagonal of Σ contains the singular values of $\Omega^\top R \Omega$. Assuming that $T = \Omega$ and comparing 5 and 7, it is possible to note that $\sigma^2 = \Sigma$.

Given that matrix R represents the correlation between the coordinates of each vector $\mathbf{u}_i$ of $\mathcal{T}$, $\forall i$, the transformation applied to R in 7 performs its diagonalization, representing R in a new orthogonal system. In this new coordinate system given by T , each coordinate j of a vector $\mathbf{u}_i$ presents maximum variance with respect to axis $\mathbf{t}_j$ and null variance with respect to the other axes. This property allows to reduce data dimension.

Therefore, using only the first k vectors of T , that is, $T_k = [\mathbf{t}_1 \mathbf{t}_2 \dots \mathbf{t}_k]$, the representation of $\mathcal{T}$ in this new coordinate system is given by:

$$\hat{\mathcal{T}}_k = T_k^\top \mathcal{T}. \quad (8)$$

In the specific case of this work, the PCA is used in two different tasks:

- to find a reduced set of k linearly and uncorrelated image descriptors, based on the *YUV* representation of the image;
- to reduce the dimension of the Vector of Locally Aggregated Descriptors (VLAD), which is described in Sect. 3.1.3.

3.1.2 Scale-invariant feature transform (SIFT)

Another technique which is used here for feature extraction is the Scale-Invariant Feature Transform, or simply SIFT. It consists of four steps [20]. :

1. Scale-space extrema detection: initially, a set of keypoints must be detected. For accomplishing such a task, the image is convolved with Gaussian filters at different scales, and the differences of successive Gaussian-blurred images are taken. Keypoints are searched as maxima/minima of the Difference of Gaussians which occur at multiple scales.
2. Keypoint localization: in this step, the candidate keypoints are localized and the unstable ones (points which are sensible to noise or with low contrast) are eliminated.
3. Orientation assignment: one or more orientations are assigned to each keypoint, based on local image gradient directions. The assigned orientations, scale and location for each keypoint enable SIFT to construct a canonical view for the keypoint, which is invariant to similarity transforms [16].
4. Keypoint descriptor: finally, keypoints are used for computing descriptor vectors.

Specifically, a keypoint descriptor used by SIFT is created by sampling the magnitudes and orientations of the image gradient in the patch around the keypoint and building orientation histograms to capture the relevant aspects of the patch. Histograms contain 8 bins each, and

each descriptor contains a 4×4 array of 16 histograms around the keypoint. This leads to a SIFT feature vector with $4 \times 4 \times 8 = 128$ elements. This 128 element vector is then normalized to unit length to enhance invariance to changes in illumination.

The main intention of SIFT-based representations is to avoid problems incurred by boundary effects [20]. Therefore, smooth changes in location, orientation and scale do not cause radical changes in the feature vector. Moreover, it is a compact representation, expressing the patch of pixels using a 128 element vector.

It should be noticed that a conversion of *RGB* images to grayscale may proceed SIFT application. It is justified by the fact that SIFT uses a one-dimensional (1D) vector of scalar values as a local feature descriptor and it cannot be extended to operate on color images¹. The main difficulty of applying SIFT to color images is that it is not possible to represent colors using 1D scalar values [8].

In this work, the SIFT is employed to find the keypoints (and the respective gradient vectors) of the fish images. Besides, it is also used jointly with VLAD and PCA to obtain smaller and fixed-length descriptors.

3.1.3 SIFT + vector of locally aggregated descriptors (VLAD + PCA)

The Vector of locally aggregated descriptors (VLAD) is a technique which is commonly employed for fitting local image descriptors, such as SIFT, into fixed-length descriptors [15]. This method aggregates the image descriptors based on its values and it delivers a fixed-length (often smaller) vector with the most important visual attributes of the input image.

Given an input image I with n descriptors, $X = [x_1, \dots, x_n]$, a VLAD can be created as follows:

1. Codebook building: a codebook, with k descriptors (or centroids), $C = [c_1, \dots, c_k]$, is built for the input image. Such as proposed by Jégou et al., this task is accomplished by a k -means clustering algorithm [24], using the n original descriptors of the image as the input.
2. Descriptor association: each descriptor x_i is associated to a centroid c_j , such that:

$$C_j = \left\{ x_i \in C_j \Leftrightarrow j = \arg \min_{j \in \{1, \dots, k\}} \|x_i - c_j\|_2, \right. \\ \left. \forall i \in \{1, \dots, n\} \right\} \quad (9)$$

in which C_j is the set of descriptors associated to centroid c_j . In this process, each descriptor is associated to the closest centroid, based on a simple Euclidean distance.

¹ Color images are generally composed of three-dimensional (3D) vector values.

3. Calculating difference vectors: each component of the difference vectors $V = [v_1, \dots, v_k]$ is calculated through the following relation:

$$v_{i,j} = \sum_{x_k \in C_i} x_{k,j} - c_{i,j} \quad (10)$$

in which:

$u_{k,i}$ is the i -th component of the vector u_k ;

x_i , x_k and c_i are $d \times 1$ vectors;

d is the number of characteristics of each original descriptor.

4. Finally, the vectors $V = [v_1, \dots, v_k]$ are L2-normalized, as shown in (11).

$$v_i \leftarrow \frac{v_i}{\|v_i\|_2} \quad (11)$$

As the main result, this technique delivers a new set of k local descriptors, and the global dimension is $D = k \times d$.

As it is possible to note from the procedure above, a VLAD is created based on the differences between the original descriptors and their respective centroids from the codebook. This procedure can be seen as an adapted and simplified version of the Fisher Kernel [29]. Besides, the employment of a codebook has been inspired from bag of features representations [35].

The main advantage of this process is to deliver a fixed-length set of local descriptors. Fixed-length representations can be compared using standard distance metrics, what makes possible to employ robust classification methods, such as neural networks, support vector machines or immune-inspired algorithms [15].

Here, the VLAD is used jointly with SIFT and PCA. Firstly, the input set of descriptors is obtained using SIFT. Then, the VLAD is employed for extracting k descriptors, with 128 elements each. Finally, the PCA is employed to reduce such a set of descriptors, to obtain a simpler, an easier to compare, characterization of the images.

3.2 Clustering

In this work, the clustering step is used as a feature compression method. Therefore, dimensionality reduction is accomplished through a two-level approach: (1) directly using specific methods for that, such as, PCA, SIFT and SIFT + VLAD + PCA, as described in the previous section and (2) indirectly using clustering techniques, such as, aiNet, ARIA and k -means, which remove training examples that are most probably not useful to the classification task.

The parameters of a detected cluster become the weighted average of the parameters of its constituent features. In this context, clustering is used as a method to extract information from the unlabelled data to boost the

classification task. That is, clustering is used as a down-sampling pre-process to classification to reduce even more the size of the training set, resulting in a less complex classification problem, which is easier and quicker to solve. After the employment of the clustering algorithm, it is expected that individuals of the same species are grouped in the same cluster, or at least close ones. Three algorithms have been proposed for performing image grouping: an artificial immune network algorithm (aiNet) [10], an adaptive radius immune algorithm (ARIA) [3] and a k -means algorithm [24].

3.2.1 Artificial immune network algorithm (aiNet)

The Artificial Immune Network, also known as aiNet, is a bio-inspired computational model which is based on the concepts of the immune network theory, mainly the interactions among B-cells (stimulation and suppression), and the cloning and mutation process [10]. It generates a network of antibodies linked according to the affinity (Euclidean distance). A subset of the best suited antibodies (with respect to a given antigen) is selected, cloned and mutated, to find better antibodies. Part of the clones is selected to be memory antibodies, by eliminating those whose affinity with the current antigen is lower than a death threshold. If a pair of memory antibodies have an affinity greater than a suppression threshold, one of them is removed from the network to avoid redundancy [10]. A basic scheme of this algorithm is shown in Algorithm 1 [10].

In this work, the aiNet is executed over the descriptors found by the feature extraction method, and it is used to build the knowledge base.

3.2.2 Adaptive radius immune algorithm (ARIA)

The adaptive radius immune algorithm, or simply ARIA, is an immune-inspired algorithm which implements an antibody adaptive suppression radius which varies inversely with the local density in the space. This feature makes it possible to maximally preserve the density of the data even in compact representations, what can be helpful in pattern recognition [3].

The ARIA can be summarized into three main phases [3]:

1. affinity maturation: the antigens are presented to the antibodies, which suffer hypermutation to better fit the antigens;
2. clonal expansion: those antibodies which are more stimulated are selected to be cloned, and;
3. network suppression: the interaction between the antibodies is quantified and if one antibody recognizes another, one of them is removed from the pool of cells.

A basic scheme of the ARIA algorithm can be seen in Algorithm 2 [3]. From this procedure, it is possible to note that the algorithm is similar to the aiNet. The most important difference between both algorithms is the employment of the adaptive radius, which tends to improve the efficiency of the algorithm.

Algorithm 1 Pseudocode for aiNet

```

1: for each iteration do
2:   for each antigen  $\mathbf{x}_i$ ,  $i = 1, \dots, p$  do
3:     determine its affinity with respect to each antibody of a random set  $\mathbf{A}$ ;
4:     select a subset  $\mathbf{A}_n$  of antibodies in  $\mathbf{A}$  which is composed of the  $n$  antibodies
       with highest affinity;
5:     clone the  $n$  selected antibodies proportionally to their antigenic affinity, gener-
       ating a set  $C$  of clones;
6:     perform a directed affinity maturation of  $C$ , generating a mutated set  $C^*$ ;
7:     determine the antigenic affinities of the mutated antibodies in  $C^*$ ;
8:     re-select  $\zeta\%$  of the antibodies in  $C^*$  with the highest affinities and put them
       into a matrix  $\mathbf{A}_m$  of clonal memory;
9:     eliminate the memory clones from  $\mathbf{A}_m$ , whose affinities are higher than a natural
       death threshold  $\sigma_d$ ;
10:    determine the affinity between pairs of memory clones;
11:    eliminate memory clones whose affinities are lower than a suppression threshold
        $\sigma_s$ ;
12:    concatenate the total antibody memory matrix with the resultant clonal mem-
       ory;
13:  end for
14:  determine the affinity between pairs of memory antibodies;
15:  eliminate all the antibodies whose affinities are lower than  $\sigma_s$ ;
16:  build the total antibody matrix;
17: end for

```

Algorithm 2 Pseudocode for ARIA

```

1: 1. Initialize variables
2: 2. For iteration 1 to  $G$  do:
3:   2.1 For each antigen  $\mathbf{x}_i$ ,  $i = 1, \dots, p$ , do:
4:     2.1.1 Select the best matching antibody  $\mathbf{A}_b$ ;
5:     2.1.2 Mutate  $\mathbf{A}_b$  with rate  $\mu$ ;
6:   end;
7:   2.2 Kill those antibodies that are not stimulated;
8:   2.3 Clone those antibodies that recognize antigens
   located at a distance larger than its radius  $R$ ;
9:   2.4 Calculate the local density for each  $\mathbf{A}_b$ ;
10:  2.5 Calculate the suppression threshold of each  $\mathbf{A}_b$ 
   making  $R_{\mathbf{A}_b} = r \times (D_e^{max}/D_e)^{(1/D_i)}$ ;
11:  2.6 Suppress antibodies giving survival priority for
   those with smaller  $R$ ;
12:  2.7 Make  $E = \text{mean}(R)$ ;
13:  2.8 If current generation is greater than  $G/2$ :
14:    2.8.1 Reduce  $\mu$  ( $\mu = \mu * \alpha$ );
15:  end;
16: end;

```

Such as for the aiNet, the set of antigens presented as input to the ARIA corresponds to the feature vectors estimated by PCA or SIFT.

3.2.3 *k*-Means

The *k*-means is an easy to implement algorithm commonly employed for clustering data [24]. It intends to divide n observations (input points) into k clusters, in such a way that each input is assigned to the cluster with the nearest center. A basic scheme for this algorithm is shown in Algorithm 3 [24].

embedded in the Euclidean space (the image descriptors), in which the Euclidean distance is defined. Besides, it should not be ignored that this algorithm has a single parameter to be set (k), what can make it easier to tune.

3.3 Classifiers

Finally, the classifiers which are employed for categorizing an input image are presented in this section. These methods are hardly dependent on the method which is used for clustering data. Three different classifiers are available: *k*-NN, SIFT classifier and *k*-means based classifier.

Algorithm 3 Pseudocode for *k*-Means

```

1: start the  $k$  centers at random;
2: while at least one center is not stable2 do
3:   for each of the  $n$  inputs do
4:     assign the input to the cluster with the nearest center;
5:   end for
6:   for each of the  $k$  clusters do
7:     set the center of the cluster with the mean of the points assigned to such a cluster;
8:   end for
9: end while

```

Despite its simplicity, the *k*-means is endowed with two drawbacks:

- it requires that a distance metric is defined for the input data;
- it is necessary to specify, a priori, the number of clusters in which the data should be split (k). Usually, this number is not known.

The limitation related to evaluation of distances is addressed here by representing the images as vectors

3.3.1 *k*-Nearest neighbor classifier (*k*-NN)

The *k*-nearest Neighbor, usually referred as *k*-NN, is a very simple method which is commonly employed for classifying objects [26]. On this method, an input is classified based on the closest training examples: the class with high occurrence amongst the k nearest neighbors of the input is elected as the input class. A scheme of this algorithm is shown in Algorithm 4 [26].

Algorithm 4 Pseudocode for k -NN

```

1: start the counters of all classes with 0;
2: for all the  $k$  nearest neighbors of the input data do
3:   increment the counter which corresponds to the class of the current neighbor;
4: end for
5: assume the class with high occurrence as the class of the input;

```

Here, a 1-NN variation of the k -NN is used to find which species shares more similarities with the input image. Therefore, when an input image is presented, the distances between the descriptor vector of such an image and all the descriptor vectors of the training images are evaluated and the species of the nearest vector is assumed for the input image. This kind of approach requires that all the descriptor vectors have equal dimension, what makes it inefficient when SIFT is employed alone.

3.3.2 SIFT classifier

The dimension of the set of image descriptors found using SIFT may vary for different images, what makes it impossible to employ simple Euclidean comparisons. Here, an incremental comparison approach has been adopted for addressing such a limitation. Given an input image I_v , this method estimates the matches between its keypoints and the keypoints of the image samples, using the match measure suggested in [3].

This classifier works as follows:

1. assume that an input image must be classified amongst t species, $i = \{1, \dots, t\}$;
2. evaluate the number of matches, m_i^j , for each pair (I_v, I_j^i) (i denotes the species considered and j denotes the image samples of the species i);
3. find the total number of matches between the descriptors of I_v and the images of each candidate species i , using

$$M^i = \sum_{j=1}^{N_i} m_j^i \quad (12)$$

in which N_i is the number of reference individuals for species i .

4. associate the I_v to the species with the highest M_i value.

This process is illustrated in Fig. 2. In this example, the classifier assigned the species 1 to the input image I_v .

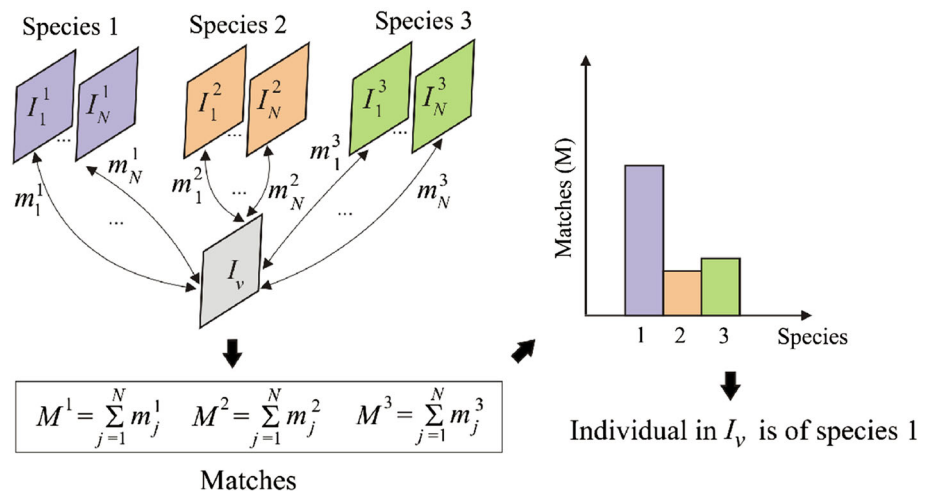
3.3.3 k -Means based classifier

A classifier which exploits the properties of k -means clustering method has been proposed here. Given that an image is to be classified, it selects a subset of m groups which are more suitable for such an input, based on the centroids obtained using the k -means. For each of these m groups, a small set of the W images which have higher similarity with the input image is returned. The species with higher occurrence amongst the returned ones is assumed as the species of the input image.

A basic scheme for this classifier is shown in Algorithm 5. In this scheme, it is assumed that an image Q is given as input and that the centroids $C = [c_1 \dots c_k]$ have been obtained previously, using k -means.

This process reduces the complexity of classifier, since it restricts the image comparisons only to a small part of the training set (the m most probable groups). It can be

Fig. 2 Operation of the SIFT classifier



Algorithm 5 Pseudocode for Relevance Feedback Classifier

```

1: for each of the  $k$  groups do
2:   evaluate the similarity between the descriptor of  $Q$  and the current centroid  $c_i$ ;
3: end for
4: for each of the  $m$  more suitable groups do
5:   return the  $W$  images with high similarity;
6: end for
7: choose the species with higher occurrence amongst the returned images;

```

specially useful for problems with large training sets, since, in those cases, the time required for performing a single query can become excessively high.

4 Datasets

Two different datasets have been considered in this work: (i) fish in formaldehyde solution, and; (ii) fish in vivo. These datasets are discussed along the two next sections.

4.1 Dataset 1: fish in formaldehyde solution

The first dataset is composed of image samples of six fish species (see Fig. 3a–f) which are perfectly conserved in formaldehyde solutions, such as considered in [25]. For each species, three different individuals have been included in the dataset. These individuals have been rotated from -40° to 40° , in 10° steps, to simulate their swimming

characteristics [25]. This rotation process results in 9 images per different individual. Since six species are evaluated, with three individuals per species and nine images per individual, the whole database is composed of 162 image samples.

This dataset has been divided into training set and evaluation set in two different ways, depending on the technique which is used for feature extraction:

Schemes 1 and 2 (PCA): in these schemes, a set of 108 images, chosen at random, has been used to estimate the knowledge base, and the remaining 54 images have been used for accuracy evaluation.

Schemes 3, 4 and 5 (SIFT and SIFT + V-LAD + PCA): for SIFT-based algorithms, the knowledge base has been created using the nine image samples of one individual for each of the six species. Therefore, the training database has been built with 54 images and the other 108 images have been used for validation.

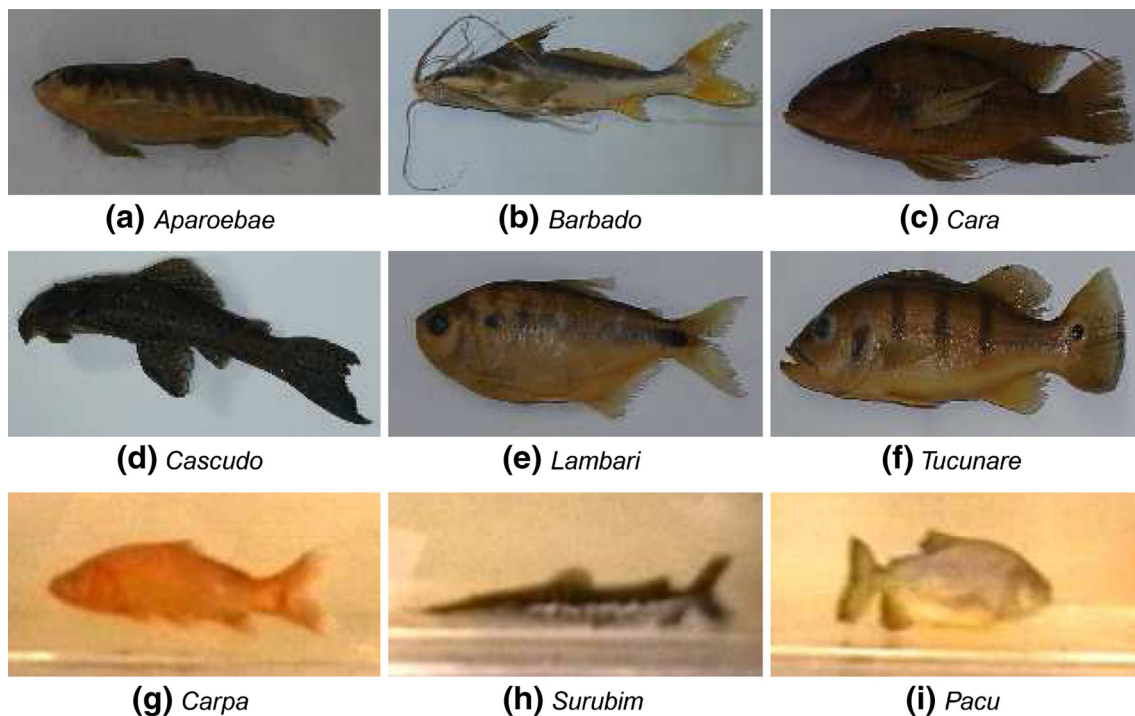


Fig. 3 Image samples of the nine fish species considered. **a–f** Fish conserved in formaldehyde solution and **g–i** fish in vivo

The evaluation of fishes in formaldehyde solution, simulating swimming rotation, has two main intentions:

1. to evaluate the accuracy of the proposed classifiers against significant variations of the 3D fish orientation;
2. to analyze the behavior of the proposed schemes in a controlled environment, without water influence.

4.2 Dataset 2: fish in vivo

The second dataset considered here is composed of images of four live fish species (Carpa, Surubim, Pacu and Cascudo), acquired at a prototype of a fish ladder (see Fig. 3g–i for some samples). In this dataset, a single individual is available for each species, and 12 images have been taken for each individual, resulting in 48 images. Such as for the first dataset, two different strategies have been used for splitting the dataset into training and evaluation sets:

Schemes 1 and 2: the 48 images have been divided, at random, into 24 images for building the knowledge base and 24 images for accuracy evaluation.

Schemes 3, 4 and 5: for these schemes, the knowledge base has been created using 6 image samples of each individual for species. Therefore, 24 images are used for finding the knowledge base and the other 24 ones are left for validation.

The main goal of this test is to demonstrate the applicability of the proposed schemes in real-world applications.

5 Results

Before presenting the results, it is important to explain how the parameters have been set in each specific module:

Feature extraction:

- PCA: the 3 first principal components are used as image descriptors
- SIFT + VLAD + PCA: the number of descriptors in the codebook has been set to 64

Clustering:

- aiNet:
 - number of generations (G): 10,
 - natural death threshold (σ_d): 1,
 - suppression threshold (σ_s): 0.01,
 - number of clone antibodies (n): 4,
 - mature antibodies to be selected (ζ): 20 %.
- ARIA:
 - number of generations (G): 10,
 - radius of each antibody (R): initially drawn in [0.01, 0.09],

- smallest radius (r): 0.01,
- mutation ratio (μ): initially set to 100 %.
- number of generations between mutation ratio updates (α): 1.

k -means:

- number of clusters (k): $\sqrt{N}$ (N : number of training samples).

Classifiers:

k -means based classifier:

- number of candidate groups (m): 3,
- number of images to be returned (W): 10,

The results which have been observed are presented in two separated sections, one for each dataset.

5.1 Results 1: fish in formaldehyde solution

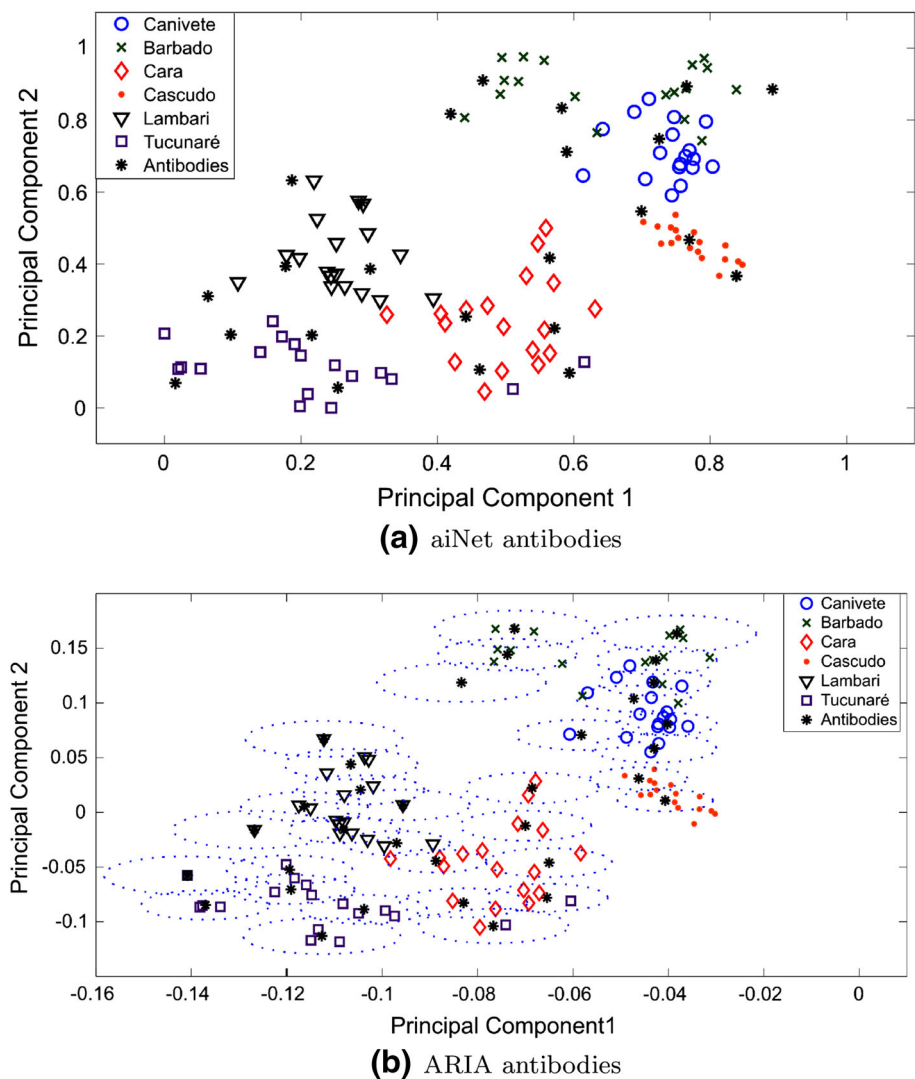
The results achieved by the five proposed schemes are shown in Table 2. From such a table, it is possible to note that the replacement of aiNet by ARIA always leads to better results: 85 % for Scheme 1 vs. 92 % for Scheme 2 and 83 % for Scheme 1 vs. 87 % for Scheme 2. This difference can be explained by the fact that ARIA is capable of capturing the relative density information in space, leading to a refined clustering result. Such a feature can be seen in Fig. 2. This figure shows a PCA feature space, composed of only two principal components, and the antibodies (center of clusters) obtained by aiNet (Fig. 4a) and ARIA (Fig. 4b). From these figures it is possible to note that ARIA could find clusters which are more suitable than the aiNet ones.

The effect of the number of principal components in the accuracy of the classifier has been also evaluated. Inside the Schemes 1 and 2, such a number has been varied from 2 to 10, as shown in Fig. 5. The figure shows that the better results are achieved for three principal components. This is in agreement with the literature [11], which shows that the average error rate is strongly related to the numbers of features and image samples. Besides, it

Table 2 Dataset 1: accuracy of the proposed schemes

Scheme	Characteristics	Accuracy (%)
1	PCA + aiNet + k -NN	85
2	PCA + ARIA + k -NN	92
3	SIFT + aiNet + SIFT class	83
4	SIFT + ARIA + SIFT class	87
5	SIFT+VLAD+PCA + k -means + k -means class	68

Fig. 4 PCA feature space for the two components with highest values, computed by applying PCA to matrix $\mathcal{Y}$. In this case, $\mathcal{Y}$ encodes color information of our image samples of fish conserved in formaldehyde solution



is possible to note that ARIA improves the accuracy of the PCA-based schemes for any number of principal components.

The impact of employing SIFT for finding image descriptors has been evaluated using Schemes 3 and 4 (examples of SIFT descriptors in fish images can be seen in Fig. 6). These methods have achieved accuracy ratios of 83 and 87 %, respectively. This reinforces the impression that ARIA is better suited than aiNet for image clustering.

An evaluation of the number of correct matches obtained by SIFT with regard to the rotation angle of the image has been also performed. A result of this evaluation can be seen in Fig. 7, for the Canivete species. This figure shows the number of matches between the image of a validation individual of the species Canivete, rotated at -40° , and all the images of the reference individual of the same species. Note that the

number of matches is higher when the rotation angles are close to -40° . When the difference between those angles increases, the number of matches decreases dramatically. This sensitivity with regard to rotation probably explains the lower performance observed for SIFT-based methods.

With regard to Scheme 5, it has shown the worst performance amongst the methods tested. The authors believe that this lower performance is possibly related to k -means. Basically, k -means clustering generates a specific number of spherical disjoint clusters of the same size. In this scenario, k -means is frequently unable to handle noise and outliers, making it not suitable to discover clusters with non-convex shapes. This is, in turn, a behavior that one may observe in our datasets, as illustrated in Fig. 4a, b. Note that the clusters are not well defined, possibly contributing to a high classification error when k -means is applied.

Fig. 5 Dataset 1: overall accuracy of the Schemes 1 and 2 as a function of the number of principal components

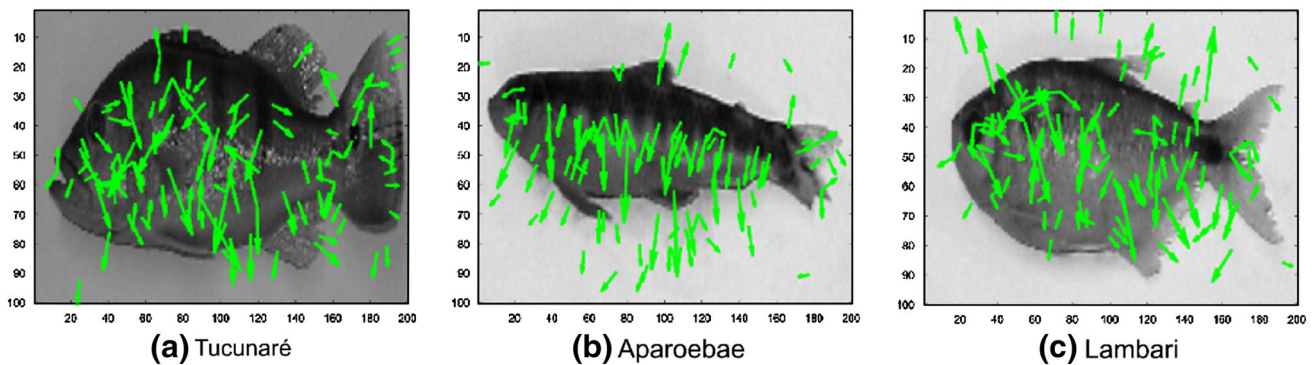
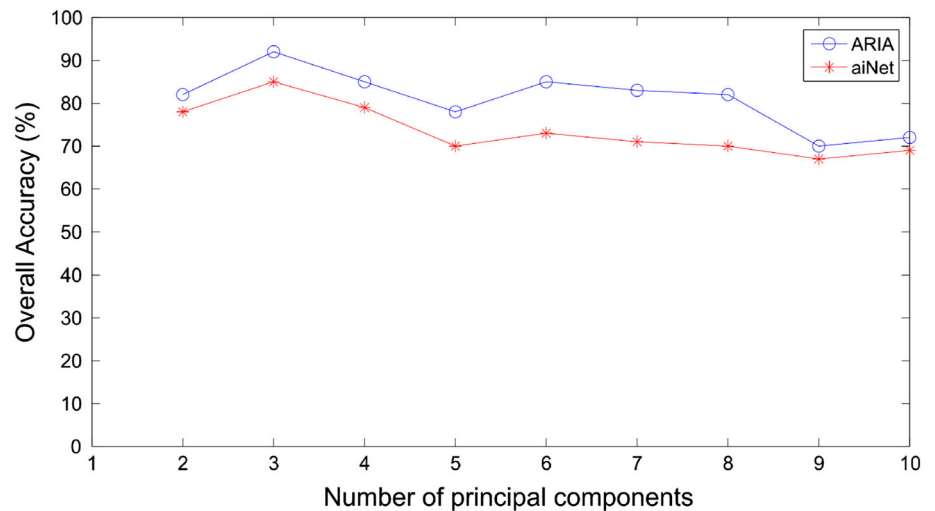
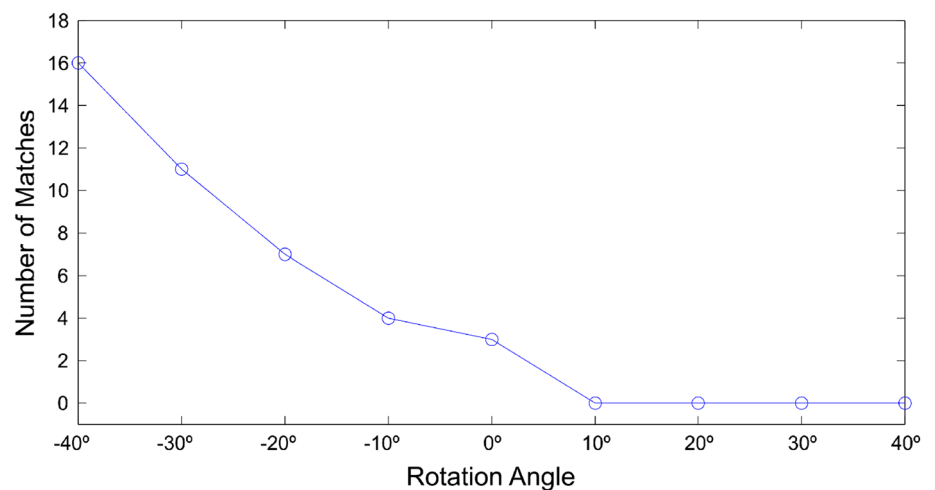


Fig. 6 Examples of keypoints extracted by SIFT algorithm in image samples

Fig. 7 Dataset 1: evaluation of robustness of SIFT with regard to image rotation—Canivete species



Comparing the results observed for the five schemes, it is possible to note that the Scheme 2, which combines PCA, ARIA and k -NN is the most adequate one for the first dataset, since it has achieved the best overall accuracy (92 %). It is followed by Schemes 1, 3, 4, and 5 in such an order.

5.2 Results 2: fish in vivo

The same methodology used in the first group of experiments has been applied to the second one. The results achieved by the five schemes in the dataset are shown in Table 3.

Table 3 Dataset 2: accuracy of the proposed schemes

Scheme	Characteristics	Accuracy (%)
1	PCA + aiNet + k -NN	92
2	PCA + ARIA + k -NN	92
3	SIFT + aiNet + SIFT class	75
4	SIFT + ARIA + SIFT class	79
5	SIFT+VLAD+PCA + k -means + k -means class	48

Such as for the first dataset, the ARIA has shown to be the most suitable choice for grouping the fish species, since it is not outperformed by aiNet in any of the schemes considered. It should be noticed that the influence of water characteristics, the arbitrary rotation and the variation in the scale of the images have not affected the performance of the PCA-based classifiers (Schemes 1 and 2). This is an important result, since it indicates a certain measure of robustness of those algorithms to aspects commonly present in real-world applications. Basically, the robustness demonstrated by the PCA-based classifiers relates to the characteristics of the image samples considered of fish in vivo. Note that, for those image samples, the influence of water characteristics, such as turbidity, make global properties such as shape and size more relevant than local ones, such as keypoints. In fact, for this dataset, the objects are weakly textured, so that fish appearance is dominated by its projected contours. Therefore, such a scenario has possibly contributed to the effectiveness of PCA, a tool commonly used for global image description. Finally, it is important to keep in mind that such a conclusion has been made based on a small dataset, and it should be verified by a larger dataset when it is available.

Besides, the decrease in the performance of the SIFT-based classifiers was not very significant in this dataset. It also suggests that these methods are hardly affected by external aspects (noise). The Scheme 5 was not robust enough for dealing with fishes in vivo.

Based on the results observed for this second dataset, it is possible to establish the following ranking for the methods: (1) Schemes 1 and 2, (3) Scheme 4, (4) Scheme 3, (5) Scheme 5. Such as expected, this ranking is very similar to the ones obtained in the first dataset, what corroborates with the conclusions taken there.

6 Conclusion

This work proposes five schemes which are intended for automatic classification of fish species. Each of these schemes is composed of a particular combination of techniques which are employed for feature extraction, data

clustering and image classification. The results achieved by the schemes in the two datasets considered (fishes in formaldehyde solution and in vivo) suggest that it is often possible to find at least a scheme with very high classification accuracy (close to 92 %). Besides, it has been observed that the Scheme 2, which combines PCA for feature extraction, ARIA for clustering and k -NN for classification, is the more suitable choice for both datasets.

A possible extension of this work is the use of one of the schemes proposed here inside a comprehensive system which automatically detects, tracks, counts and classifies fish in more representative underwater video datasets. Moreover, motivated by the lack of a systematic comparison of dimensionality reduction techniques, we plan to perform a comparative study between well-known linear dimensionality reduction techniques, such as PCA, and nonlinear dimensionality reduction techniques, such as those proposed in [1, 34, 36]. The aims of this study will be to investigate to what extent nonlinear dimensionality reduction techniques outperform the traditional PCA on real-world datasets of fish species and to identify the inherent weaknesses of the nonlinear techniques for dimensionality reduction. Finally, we have in mind to investigate to what extent NMF algorithm [13, 18, 21–23, 37] outperforms the traditional K-means, the immune algorithms aiNet and ARIA and eventually other state-of-art clustering methods on real-world datasets of fish species, as well as providing a deep understanding of the sensitivity of the NMF method to outliers.

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